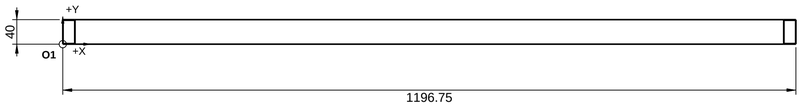


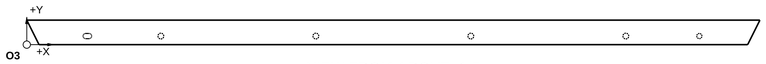

**VIEW FROM BELOW 1:5**

X = +X, Y = -Y (pod axes); looking up (toward +Z); origin O2




**PRINCIPAL VIEW 1:5**

X = +X, Y = +Z (pod axes); looking toward +Y; origin O1




**VIEW FROM ABOVE 1:5**

X = +X, Y = +Y (pod axes); looking down (toward -Z); origin O3

**HOLES H01-H06: C1 BACK MEMBER 40X40, bottom face**
seen in the view from below; origin O2; +X right, +Y up; drilled from this face

| TAG | X (mm) | Y (mm) | SIZE | DEPTH | FEATURE (register) AND FITS |
|---|---|---|---|---|---|
| H01 | 98.88 | -14.00 | SLOT 8.5 x 14 OVERALL, R4.25 ENDS along X | THRU 1 WALL (1) | F-4111-10 Spigot hole and slot |
| H02 | 218.88 | -14.00 | Ø9.53 | THRU 1 WALL (1) | F-4111-09 Back rivnut holes, J-CEIL-01 |
| H03 | 471.88 | -14.00 | Ø9.53 | THRU 1 WALL (1) | F-4111-09 Back rivnut holes, J-CEIL-01 |
| H04 | 724.88 | -14.00 | Ø9.53 | THRU 1 WALL (1) | F-4111-09 Back rivnut holes, J-CEIL-01 |
| H05 | 977.87 | -14.00 | Ø9.53 | THRU 1 WALL (1) | F-4111-09 Back rivnut holes, J-CEIL-01 |
| H06 | 1097.87 | -14.00 | Ø8.5 | THRU 1 WALL (1) | F-4111-10 Spigot hole and slot |

| | | | |
|---|---|---|---|
| PROJECT: **Spaceship (FocusPod v2)** | STATUS: **DRAFT** | DATE: 2026-11-01 | SHEET: **1/2** |

TITLE
**CEILING - FRAME BACK MEMBER 40 X 40 - C1**
Trần trái - thanh sau 40x40 của khung

| PART NUMBER | REV | DRAWING NUMBER (FILE) |
|---|---|---|
| **412104** | **P01** | **412104_P01_CEILING-FRAME-BACK-MEMBER-40-X-40-C1_DWG** |

MATERIAL
MATERIAL: Galvanized steel S280GD+Z or SGC400 (yield 280 MPa or more), Z100

FINISH
FINISH: pre-galvanized Z100; each weld's burnt zone wire-brushed and coated with 2-component zinc-rich epoxy (J-FRAME-01); the bond prep on the skin face (D-137)

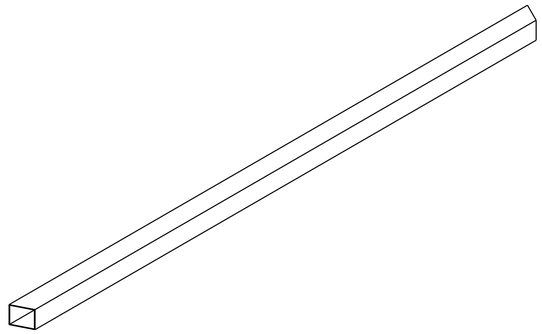
| SCALE | SIZE | UNITS | MASS |
|---|---|---|---|
| 1:5 | A1 | **UNITS: mm** | 1.438 kg (calculation) |

| CREATED BY | APPROVED | TOLERANCES |
|---|---|---|
| Autonomous | not released | TOLERANCE: see the notes; not ISO 2768 |

PROJECTION
ISO first angle (ISO 5456-2)

## GENERAL NOTES

1. UNITS: mm. Do not scale the drawing. ISO first-angle projection (ISO 5456-2), symbol in the title block. Dimensions to ISO 129-1. Sheet 1: views, sections and hole tables; sheet 2: these general notes.
2. Views: the principal view looks toward +Y (X = +X, Y = +Z (pod axes); looking toward +Y); every view's title gives its own X and Y and how they map to the pod axes (conventions section 4). On every view X runs right and Y up.
3. Origin O = the part's corner shown; O1, O2... are O seen in each view. Hole positions in the tables are from O.
4. Cut from pre-galvanized steel tube (S280GD+Z or SGC400, Z100) to the length and end cuts shown, as the frame's cut list row gives them (D-120, D-131); each mitred end cut back square to its mitre for the 2.0 mitre plate (J-FRAME-01). Welded into its frame (411101 or 411102) in the frame's jig.
5. Holes, where the tables show them: made after welding and the zinc-rich touch-up, by CNC or a drill jig that locates on the frame's joint faces and its spigot hole (J-FRAME-01); this sheet shows them on the tube for reference. Rivnuts 838003 are set after that.
6. Holes go THRU 1 WALL of the member unless the table says otherwise; deburr both sides.
7. TOLERANCES (PROPOSED until the shop confirms; not ISO 2768): cut length +0/-0.5 and end angles ±0.25° (the cut list, assumption); hole positions to the frame's datums (R-J6: Ø0.4 to A|B|C, the spigot hole and slot ±0.1); rivnut holes Ø9.53 +0.10/0 (maker); other hole sizes +0.2/0 (assumption). Not ISO 2768 on the tube.
8. Source: v2/03-Manufacturing/ceiling/410000_P01_CEILING_STEP.step (sha256 c9d36e8963f4; declared unit cm, read in mm and checked against tools/checks/stepfile.py, worst box difference 0 mm); parts list v2/03-Manufacturing/ceiling/410000_P01_CEILING-PARTS-LIST_BOM.csv; feature rationale v2/02-Drawings/ceiling/FEATURE-RATIONALE.md. From that STEP: regenerate it, never edit the PDF.



**ISOMETRIC VIEW 1:5**
for identification only

| | | | |
|---|---|---|---|
| PROJECT: **Spaceship (FocusPod v2)** | STATUS: **DRAFT** | DATE: 2026-11-01 | SHEET: **2/2** |

TITLE
**CEILING - FRAME BACK MEMBER 40 X 40 - C1**
Trần trái - thanh sau 40x40 của khung

| | | |
|---|---|---|
| PART NUMBER: **412104** | REV: **P01** | DRAWING NUMBER (FILE): **412104_P01_CEILING-FRAME-BACK-MEMBER-40-X-40-C1_DWG** |

MATERIAL
MATERIAL: Galvanized steel S280GD+Z or SGC400 (yield 280 MPa or more), Z100

FINISH
FINISH: pre-galvanized Z100; each weld's burnt zone wire-brushed and coated with 2-component zinc-rich epoxy (J-FRAME-01); the bond prep on the skin face (D-137)

| | | | |
|---|---|---|---|
| SCALE: 1:5 | SIZE: A1 | UNITS: **UNITS: mm** | MASS: 1.438 kg (calculation) |
| CREATED BY: Autonomous | APPROVED: not released | TOLERANCES: TOLERANCE: see the notes; not ISO 2768 | |

PROJECTION: ISO first angle (ISO 5456-2)